
MHF 4U1

CHAPTER 4: TRIGONOMETRY

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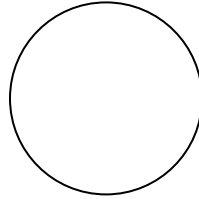
CHAPTER 4 TEST DATE:

4.1 RADIAN MEASURE

Angles are commonly measured in degrees. In mathematics and physics, however, there are many applications in which expressing the size of an angle as a pure number, without units, is more convenient than using degrees.

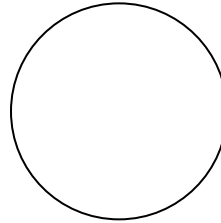
What is a radian?

The radian measure of an angle θ is defined as the length a , of the arc that subtends the angle divided by the radius, r , of the circle.

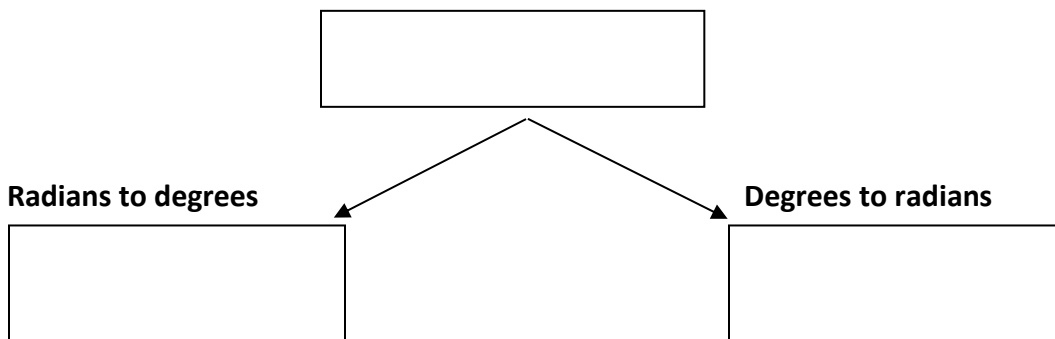


Ex.# 1: An arc of a circle measuring 22.5 cm subtends a central angle of $\frac{4\pi}{3}$ rad. Find the exact and approximate radius of the circle.

Ex.# 2: Find the measure of θ for one complete revolution.



You can use this relationship to convert between degrees and radians:



Ex.# 3: Convert to radian measure in exact form.

a) 45°

b) 270°

Ex.# 4: Convert to radian measure in approximate form.

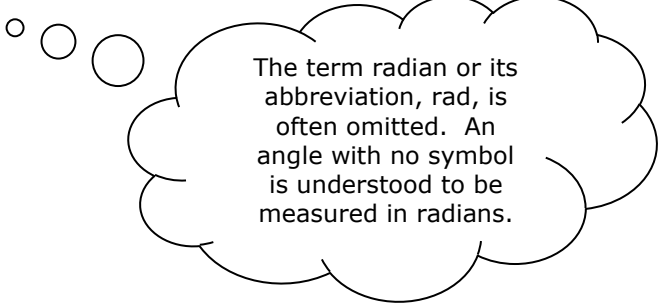
a) 202.5°

b) 9°

Ex.# 5: Convert each radian measure to degrees.

a) $\frac{5\pi}{6}$ rad

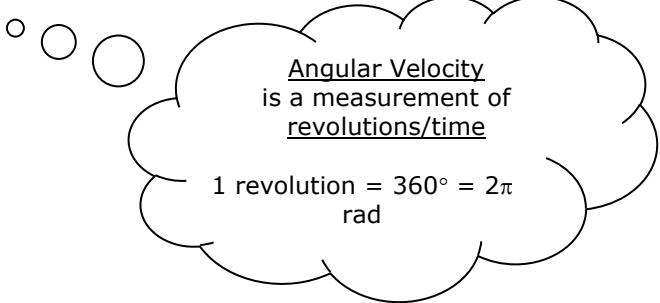
b) 1.75



The term radian or its abbreviation, rad, is often omitted. An angle with no symbol is understood to be measured in radians.

Ex. #6: The London Eye Ferris wheel has a diameter of 135m and completes one revolution in 30 minutes.

a) Determine the angular velocity in radians/second.



Angular Velocity
is a measurement of
revolutions/time

1 revolution = $360^\circ = 2\pi$
rad

b) How far has a rider travelled at 10 minutes into the ride?

Try it now!



1. The minute hand of a clock measure 8 cm from tip to pivot point.
 - a. What angle does the minute hand go through as it moves from 9:15 AM to 9:51 AM? Express your answer in radian measure.
 - b. What exact distance does the tip of the minute hand travel during this period of time?

4.2 TRIGONOMETRIC RATIOS AND SPECIAL ANGLES

Ex. 1: State the **approximate** decimal value of the following, to four decimal places.

a) $\cos 100^\circ =$

b) $\csc 200^\circ =$

Given any angle in radian measure, you can use a calculator to calculate the trigonometric ratios by setting the angle mode to radians (RAD), instead of degrees (DEG).

Ex. 2: State the **approximate** decimal value of the following, to four decimal places.

a) $\sin \frac{5\pi}{8} \text{ rad}$

b) $\csc \frac{4\pi}{3}$

c) $\cot 0.51$

Recall (from grade 11):

The 5 main ideas to remember when determining exact values of trigonometric ratios are:

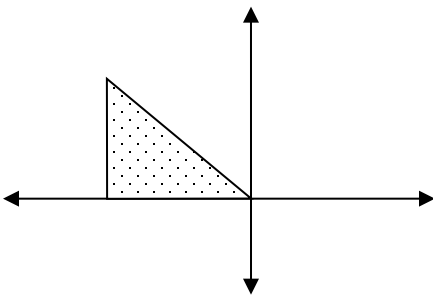
1. Given a right-angled triangle, we can solve for lengths and angles using SOHCAHTOA.

2. STANDARD POSITION

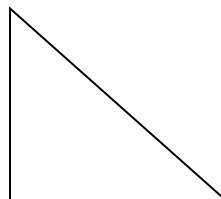
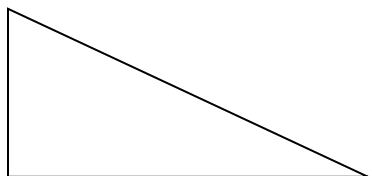
Using the Pythagorean Theorem,
 $x^2 + y^2 = r^2$, $r > 0$, we can identify the following:

Primary Trig Ratios

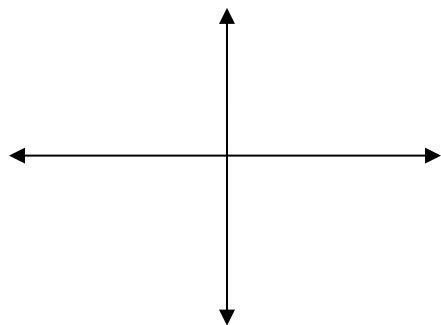
Reciprocal Trig Ratios



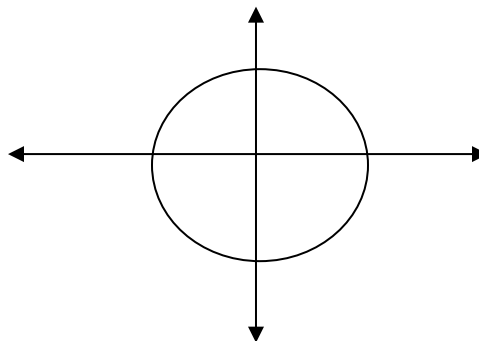
3. SPECIAL TRIANGLES



4. THE CAST RULE



5. THE UNIT CIRCLE



Ex. 3: Determine the exact values of the six trigonometric ratios for the following angles.

a) π

b) $\frac{5\pi}{6}$

4.2 DAY 2: APPLICATIONS OF TRIGONOMETRIC RATIOS AND SPECIAL ANGLES

Ex. 1: Determine an exact value for the expression:

$$\frac{\sin \frac{\pi}{6} \tan \frac{5\pi}{3}}{\csc \frac{3\pi}{4}}$$

Ex. 2: Ray is flying his kite at the end of a 50 m string. The sun is directly overhead, and the string makes an angle of $\frac{\pi}{6}$ with the ground. The wind speed increases, and the kite flies higher until the string makes an angle of $\frac{\pi}{3}$ with the ground. Determine the exact and approximate horizontal distance that the shadow moves between the two positions of the kite.

Try it now!

1. Determine the exact value of the following in standard position.

a) $\sin(-\frac{3\pi}{4})$

b) $\cot(\frac{11\pi}{6})$

2. Solve for A given that $2\sin^2(A) - \sin(A) - 1 = 0$ and $0 \leq A \leq 2\pi$.

Try it now!

1. Determine the principal angle for the following angles.

a) $-\frac{5\pi}{4}$ _____

b) $\frac{13\pi}{6}$ _____

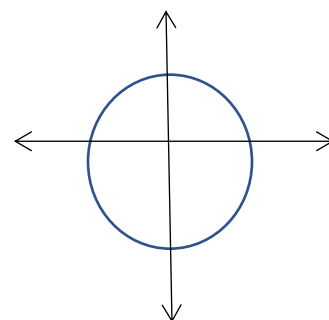
2. The point $P(\frac{1}{2}, \frac{\sqrt{3}}{2})$ lies on the unit circle.

a) Sketch the unit circle showing the point P and the standard position angle, θ , drawn to the terminal arm containing the point P.

b) Determine the exact values of the six trigonometric ratios for θ .

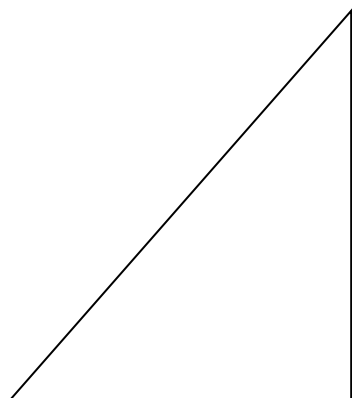
c) Using one of the trigonometric ratios from b), calculate the measure of angle θ in both degrees and radians.

d) Using the sketch from a), identify the coordinates of the point on the unit circle that is also on the terminal arm of standard position angle $\pi - \theta$.



4.3 EQUIVALENT TRIGONOMETRIC EXPRESSIONS

Let's examine the general case right-angled triangle to complete the following ratios. Recall: $90^\circ = \frac{\pi}{2}$ rad; so they can be used interchangeably.



$$\sin \theta =$$

$$\sin (90^\circ - \theta) =$$

$$\cos \theta =$$

$$\cos (90^\circ - \theta) =$$

$$\tan \theta =$$

$$\tan (90^\circ - \theta) =$$

$$\csc \theta =$$

$$\csc (90^\circ - \theta) =$$

$$\sec \theta =$$

$$\sec (90^\circ - \theta) =$$

$$\cot \theta =$$

$$\cot (90^\circ - \theta) =$$

Let's summarize our findings:

Ex#1. Use an equivalent identity to find the exact value of $\csc 75^\circ$ given that $\sec 15^\circ = \sqrt{6} - \sqrt{2}$.

Recall: the two acute angles in the above triangle are complementary angles (have a sum of 90°). All complementary angles (degrees or radians) share the same properties. Expressing the above angles in radian measure gives the following relationships:

Trigonometric Co-function Identities of $\frac{\pi}{2}$

$\sin \theta = \cos \left(\frac{\pi}{2} - \theta \right)$	$\sin \left(\theta + \frac{\pi}{2} \right) = \cos \theta$
$\cos \theta = \sin \left(\frac{\pi}{2} - \theta \right)$	$\cos \left(\theta + \frac{\pi}{2} \right) = -\sin \theta$
$\tan \theta = \cot \left(\frac{\pi}{2} - \theta \right)$	$\tan \left(\theta + \frac{\pi}{2} \right) = -\cot \theta$
$\csc \theta = \sec \left(\frac{\pi}{2} - \theta \right)$	$\csc \left(\theta + \frac{\pi}{2} \right) = \sec \theta$
$\sec \theta = \csc \left(\frac{\pi}{2} - \theta \right)$	$\sec \left(\theta + \frac{\pi}{2} \right) = -\csc \theta$
$\cot \theta = \tan \left(\frac{\pi}{2} - \theta \right)$	$\cot \left(\theta + \frac{\pi}{2} \right) = -\tan \theta$
QUADRANT I	QUADRANT II

Ex#2. Use equivalent trigonometric expressions to show that $\sin \frac{\pi}{5} = \cos \frac{3\pi}{10}$.

Ex#3. Use equivalent trigonometric expressions to show that $\cos \frac{2\pi}{9} = \sin \frac{13\pi}{18}$.

Ex#4. Given that $\sin \frac{2\pi}{3} = \cos y$, first express $\frac{2\pi}{3}$ as a sum of $\frac{\pi}{2}$ and an angle, then determine the measure of angle y , and evaluate $\cos y$ to 4 decimal places.

Ex#5. Given that $\sec \frac{4\pi}{21} = 1.21$, use equivalent trigonometric expressions to evaluate the following, to two decimal places.

a) $\csc \frac{13\pi}{42}$

b) $\csc \frac{29\pi}{42}$

4.4 COMPOUND ANGLE FORMULAE

- A trigonometric expression that depends on two or more angles is known as a compound-angle expression.
- You can develop compound-angles formulas using algebra and the unit circle.
- You can apply compound angle formulas to determine exact trigonometric ratios for angles that can be expressed as sums or differences of special angles.

The Compound Angle, or Addition and Subtraction Formulas for Sine and Cosine are:

$\sin(x + y) = \sin x \cos y + \cos x \sin y$
$\sin(x - y) = \sin x \cos y - \cos x \sin y$
$\cos(x + y) = \cos x \cos y - \sin x \sin y$
$\cos(x - y) = \cos x \cos y + \sin x \sin y$

Ex. 1: Determine the exact value of each expression.

a) $\cos 15^\circ$

b) $\sin \frac{11\pi}{12}$

- **Compound angle formulas can be used, both forwards and backwards, to evaluate and simplify trigonometric expressions.**

Ex. 2: Use the appropriate compound angle formula to express each of the following as a single trigonometric function, and then determine an exact value for each.

a) $\sin \frac{\pi}{2} \cos \frac{\pi}{4} - \cos \frac{\pi}{2} \sin \frac{\pi}{4}$

b) $\cos \frac{\pi}{12} \cos \frac{7\pi}{4} + \sin \frac{\pi}{12} \sin \frac{7\pi}{4}$

Ex. 3: Evaluate $\sin(x + y)$, where x and y are obtuse angles, given: $\sin x = \frac{3}{5}$ and $\sin y = \frac{5}{13}$.

Try it now!

Simplify: $\sin\left(\frac{\pi}{2} - x\right) + \sin\left(\frac{\pi}{2} + x\right) - 2\cos x$

4.5 PROVING TRIGONOMETRIC IDENTITIES

A trigonometric identity is an equation that is true for all angles in the domain of the expression on both sides. To prove that a given equation is an identity, the two sides of the equation must be shown to be equivalent. This can be accomplished by using a variety of strategies, such as:

- ❖ Separate LS from RS
- ❖ Simplify the more complicated side until it is identical to the other side
- ❖ Rewrite expressions in terms of $\sin x$ and $\cos x$ (i.e. $\tan x = \sin x / \cos x$; $\csc^2 x = 1 / \sin^2 x$)
- ❖ Rewrite expressions using any of the identities in the tables provided
- ❖ Find a common denominator to combine fractions
- ❖ Factor where possible
- ❖ Multiply by the conjugate binomial to simplify denominators with a sum or difference
- ❖ Do not skip steps
- ❖ Do not cross multiply
- ❖ End your proof with **LS = RS //**

Cofunction Identities:

Compound Angle Formulas:

Quadrant 1		Quadrant 2	
$\sin x = \cos\left(\frac{\pi}{2} - x\right)$	$\cos x = \sin\left(\frac{\pi}{2} - x\right)$	$\sin\left(x + \frac{\pi}{2}\right) = \cos x$	$\cos\left(x + \frac{\pi}{2}\right) = -\sin x$
$\tan x = \cot\left(\frac{\pi}{2} - x\right)$	$\cot x = \tan\left(\frac{\pi}{2} - x\right)$	$\tan\left(x + \frac{\pi}{2}\right) = -\cot x$	$\cot\left(x + \frac{\pi}{2}\right) = -\tan x$
$\csc x = \sec\left(\frac{\pi}{2} - x\right)$	$\sec x = \csc\left(\frac{\pi}{2} - x\right)$	$\csc\left(x + \frac{\pi}{2}\right) = \sec x$	$\sec\left(x + \frac{\pi}{2}\right) = -\csc x$

$\sin(x + y) = \sin x \cos y + \cos x \sin y$
$\sin(x - y) = \sin x \cos y - \cos x \sin y$
$\cos(x + y) = \cos x \cos y - \sin x \sin y$
$\cos(x - y) = \cos x \cos y + \sin x \sin y$
$\sin 2x = 2 \sin x \cos x$
$\cos 2x = \cos^2 x - \sin^2 x$

Identities from MCR 3U1 that you

are required to know:

$\tan x = \frac{\sin x}{\cos x}$	$\csc x = \frac{1}{\sin x}$	$\sec x = \frac{1}{\cos x}$	$\cot x = \frac{1}{\tan x}$	$\sin^2 x + \cos^2 x = 1$	$1 + \tan^2 x = \sec^2 x$	$1 + \cot^2 x = \csc^2 x$
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Prove: a) $\sin^2 x + \cos^2 x + \tan^2 x = \sec^2 x$

Proof:

$$RS = \sec^2 x \qquad LS = \sin^2 x + \cos^2 x + \tan^2 x$$

$$= 1 + \tan^2 x$$

$$= \sec^2 x$$

$$\therefore LS = RS //$$

b) $\frac{\sin \theta}{\csc \theta} + \frac{\cos \theta}{\sec \theta} = 1$

Proof:

$$RS = 1 \qquad LS = \frac{\sin \theta}{\csc \theta} + \frac{\cos \theta}{\sec \theta}$$

$$= \frac{\sin \theta}{\frac{1}{\sin \theta}} + \frac{\cos \theta}{\frac{1}{\cos \theta}}$$

$$= \sin^2 \theta + \cos^2 \theta$$

$$= 1$$

$$\therefore LS = RS //$$

c) $\sin 2x = \tan x(1 + \cos 2x)$

d) $\frac{\cos^2 x - \sin^2 x}{\cos^2 x + \sin x \cos x} = 1 - \tan x$

$$\text{e) } \frac{1 - \cos \theta}{\sec \theta - 1} = \cos \theta$$

$$\text{f) } \frac{1 - \sin x}{\cos x} = \frac{\cos x}{1 + \sin x}$$

g) $\cos(x+y)\cos(x-y) = \cos^2 x + \cos^2 y - 1$

EXTRA PRACTICE! TRY THESE ON YOUR OWN. PROVE EACH IDENTITY.

a) $\sec x - \cos x = \sin x \tan x$

b) $\cos^2 x - \sin^2 x = 1 - 2\sin^2 x$

c) $\frac{-\sin \theta}{\cos \theta + 1} = \frac{\cos \theta - 1}{\sin \theta}$

d) $\frac{1 - \sin \theta}{\cos \theta} = \frac{\cos \theta}{1 + \sin \theta}$

e) $\frac{\sec \theta - 1}{1 - \cos \theta} = \sec \theta$

f) $\frac{1 + \tan \theta}{\sin \theta} - \sec \theta = \csc \theta$

g) $\tan x + \cot x = \frac{1}{\cos x \sin x}$

h) $\frac{\sin \theta - \cos \theta}{\cos \theta} + \frac{\sin \theta + \cos \theta}{\sin \theta} = \sec \theta \csc \theta$

i) $\frac{\sqrt{1 - \cos^2 t} \times \sqrt{\sec^2 t - 1}}{\cos t} = \tan^2 t$

j) $\frac{1 + \sin x}{1 + \csc x} + \frac{1 + \cos x}{1 + \sec x} = \sin x + \cos x$

k) $\frac{1 + \tan^2 t}{1 + \cot^2 t} = \frac{1 - \cos^2 t}{\cos^2 t}$

l) $\frac{\sin 2x}{1 - \cos 2x} = \cot x$

m) $\frac{1}{1 + \sec \theta} + \frac{1}{1 - \sec \theta} = -2 \cot^2 \theta$

n) $\frac{\cos 2x + 1}{\sin 2x} = \cot x$

o) $\tan x + \cot x = \sec^2 x \cdot \cot x$

p) $(\sin x + \cos x)^2 = 1 + \sin 2x$

q) $\cos^4 \theta - \sin^4 \theta = \cos 2\theta$

r) $\cot \theta + \tan \theta = 2 \csc 2\theta$

s) $\frac{2 \tan \theta}{1 + \tan^2 \theta} = \sin 2\theta$

t) $\sec 2t = \frac{\csc t}{\csc t - 2 \sin t}$

u) $\csc 2a = \frac{1}{2} \sec a \csc a$

v) $\cos^4 \theta + \sin^4 \theta - \frac{1}{2} \sin^2 2\theta = \cos^2 2\theta$

w) $\sec \theta = \frac{\sin 2\theta}{\sin \theta} - \frac{\cos 2\theta}{\cos \theta}$

x) $\csc^2 x = \frac{2 \sec 2x}{\sec 2x - 1}$

y) $4 \sin^2 t \cos^2 t + \cos^2 2t = 1$

z) $1 - \cos^4 \theta = 2 \sin^2 \theta - \sin^4 \theta$

CHAPTER 4 EXTRA REVIEW

1. An arc 33 m long subtends a central angle of a circle with a radius of 16 m. Determine the measure of the central angle in radians.
2. A circle has a radius of 75 cm and a central angle of $\frac{14\pi}{15}$. Determine the arc length.
3. Convert each of the following to exact radian measure.
 - a) 20°
 - b) -50°
 - c) 160°
 - d) 420°
4. Convert each of the following to degree measure.
 - a) $\frac{\pi}{4}$
 - b) $\frac{-5\pi}{4}$
 - c) $\frac{8\pi}{3}$
 - d) $\frac{-2\pi}{3}$
5. For each of the following values of $\sin \theta$, determine the measure of θ if $\frac{\pi}{2} \leq \theta \leq \frac{3\pi}{2}$.
 - a) $\frac{1}{2}$
 - b) $\frac{-\sqrt{3}}{2}$
 - c) $\frac{\sqrt{2}}{2}$
 - d) $\frac{-1}{2}$
6. If $\cos \theta = \frac{-5}{13}$ and $0 \leq \theta \leq 2\pi$, determine
 - a) $\tan \theta$
 - b) $\sec \theta$
 - c) the possible values of θ to the nearest tenth.
7. Which of the following pairs of angles are equivalent?
 - a) 20° and $\frac{\pi}{9}$
 - b) -270° and $\frac{-3\pi}{2}$
 - c) 135° and $\frac{3\pi}{4}$
 - d) all of the above
8. The point $(-4, 7)$ lies on the terminal arm of the angle θ . What is the measure of θ in radians?
 - a) 4.19
 - b) 119.74
 - c) 2.09
 - d) 2.62
9. If $\sin \theta = \frac{-\sqrt{3}}{2}$, what are the possible values of $\cos \theta$ and $\tan \theta$?
 - a) $\cos \theta = \frac{1}{2}$, $\tan \theta = -\sqrt{3}$
 - b) $\cos \theta = \frac{-1}{2}$, $\tan \theta = -\sqrt{3}$
 - c) $\cos \theta = \frac{-1}{2}$, $\tan \theta = \frac{1}{\sqrt{3}}$
 - d) $\cos \theta = \frac{-1}{2}$, $\tan \theta = \frac{1}{\sqrt{3}}$
10. Which of the values of x , where $x \in [0, 2\pi]$, satisfy $\sin x = 0.5$?
 - a) $\frac{\pi}{6}$ and $\frac{7\pi}{6}$
 - b) $\frac{\pi}{3}$ and $\frac{5\pi}{3}$
 - c) $\frac{\pi}{6}$ and $\frac{11\pi}{6}$
 - d) $\frac{\pi}{6}$ and $\frac{5\pi}{6}$
11. Use a compound angle formula to determine a trigonometric expression that is equivalent to the following expression.
$$\sin \left(x - \frac{4\pi}{3}\right)$$
12. Without the use of a calculator evaluate exactly: $\cos \frac{\pi}{9} \cos \frac{19\pi}{18} - \sin \frac{\pi}{9} \sin \frac{19\pi}{18}$
13. Simplify each expression without the use of a calculator:
 - a) $2\sin \frac{\pi}{12} \cos \frac{\pi}{12}$
 - b) $\cos^2 \frac{\pi}{12} - \sin^2 \frac{\pi}{12}$

14. Determine the values of $\sin 2x$, and $\cos 2x$ given:

a) $\cos x = \frac{3}{5}$ and x is acute

b) $\sin x = \frac{-12}{13}$ and $\frac{3\pi}{2} \leq x \leq 2\pi$

15. Determine whether each of the following is a trigonometric equation or a trigonometric identity.

a) $\sec^2 x - \tan^2 x = \cos x$

b) $\csc^2 x - \cot^2 x = \sin^2 x + \cos^2 x$

c) $\tan^2 x = 1$

16. Prove that $\frac{1 - \sin^2 x}{\cot^2 x} = 1 - \cos^2 x$ is a trigonometric identity.

17. Prove that $\frac{1 - 2\sin^2 x}{\cos x + \sin x} + 2\sin \frac{x}{2} \cos \frac{x}{2} = \cos x$

18. Determine the solution(s) for each of the following equations, where $0 \leq x \leq 2\pi$,

a) $\cos x = \frac{\sqrt{3}}{2}$

b) $\tan x = -\sqrt{3}$

c) $\sin x = \frac{-\sqrt{2}}{2}$

19. The angle lies in the interval $\frac{\pi}{2} \leq \theta \leq \pi$ and $\sin^2 x = \frac{8}{9}$. Without using a calculator, determine the value of each of the following:

a) $\sin 2x$

b) $\cos 2x$

20. Which of these is an equivalent trigonometric ratio for $\sin \frac{2\pi}{5}$?

a) $\cos \frac{\pi}{10}$

b) $\sin \frac{3\pi}{5}$

c) $-\cos \frac{9\pi}{10}$

d) all of these

CHAPTER 4 EXTRA REVIEW ANSWERS

1. $\frac{33}{16}$

2. 70π

3.a) $\frac{\pi}{9}$

b) $\frac{-5\pi}{18}$

c) $\frac{8\pi}{9}$

d) $\frac{7\pi}{3}$

4.a) 45°

b) -225°

c) 480°

d) -120°

5.a) $\frac{5\pi}{6}$

b) $\frac{4\pi}{3}$

c) $\frac{3\pi}{4}$

d) $\frac{7\pi}{6}$

6.a) $\tan \theta = \frac{12}{5}, -\frac{12}{5}$

b) $\sec \theta = \frac{-13}{5}$

c) about 1.97 or 4.32

7. d

8. c

9. a

10. d

11. $\frac{\sqrt{3}}{2} \cos x - \frac{1}{2} \sin x$

12. $\frac{-\sqrt{3}}{2}$

13. a) $\frac{1}{2}$

b) $\frac{\sqrt{3}}{2}$

14. a) $\sin 2x = \frac{24}{25}, \cos 2x = -\frac{7}{25}$

b) $\sin 2x = \frac{-120}{169}, \cos 2x = -\frac{119}{169}$

15 a) trig eqn

b) trig identity

c) trig eqn

18. a) $x = \frac{\pi}{6}$ or $x = \frac{11\pi}{6}$

b) $x = \frac{2\pi}{3}$ or $x = \frac{5\pi}{3}$

c) $x = \frac{5\pi}{4}$ or $x = \frac{7\pi}{4}$

19. a) $\frac{-4\sqrt{2}}{9}$

b) $\frac{-7}{9}$

20. d

CHAPTER 4 : MULTIPLE CHOICE QUESTIONS

1. Determine the exact value of $\sin \frac{\pi}{6}$.

a. $\frac{2}{\sqrt{3}}$

b. 2

c. $\frac{\sqrt{3}}{2}$

d. $\frac{1}{2}$

2. Determine the exact value of $\csc \frac{\pi}{3}$.

a. 2

b. $\sqrt{3}$

c. $\frac{\sqrt{3}}{2}$

d. $\frac{2}{\sqrt{3}}$

3. Determine the exact value of $\sec \frac{7\pi}{6}$.

a. $-\frac{2}{\sqrt{3}}$

b. $\frac{\sqrt{3}}{2}$

c. $-\frac{\sqrt{3}}{2}$

d. $\frac{1}{\sqrt{3}}$

4. Determine the exact value of $\cot \frac{5\pi}{3}$.

a. $\frac{1}{\sqrt{3}}$

b. $\sqrt{3}$

c. $-\frac{1}{\sqrt{3}}$

d. $\frac{1}{2}$

5. Determine the exact value of $\csc \frac{\pi}{2}$.

a. 0

b. -1

c. 1

d. undefined

6. An equivalent trigonometric expression for $\cos\left(\frac{\pi}{2} - x\right)$ is

a. $\sin x$

b. $\tan x$

c. $\cos x$

d. none of the above

7. An equivalent trigonometric expression for $\tan(-x)$ is

a. $\tan x$

b. $-\tan x$

c. $\cot x$

d. $-\cot x$

8. Which compound angle expression is equivalent to $\sin x \cos y + \cos x \sin y$?

a. $\sin(x + y)$

b. $\cos(x + y)$

c. $\sin(x - y)$

d. $\cos(x - y)$

9. Which compound angle expression is equivalent to $\cos x \cos y - \sin x \sin y$?

a. $\sin(x + y)$

b. $\cos(x + y)$

c. $\sin(x - y)$

d. $\cos(x - y)$

Try it now!

Prove the following identity. $\frac{\sin 2x + \cos 2x + 1}{\sin 2x - \cos 2x + 1} = \cot x$